Mathematical Formulations and Algorithmic Analysis

of the Autonomous Mobile Robot (AMR) Project

Full Analytical Derivations: Kinematics, Dynamics, EKF, AMCL, Graph SLAM, MPPI, and Evasion Geometry

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Abstract

This document delivers an exhaustive mathematical and algorithmic breakdown of the Autonomous Mobile Robot (AMR) platform. The analysis covers every governing equation implemented in the project codebase, spanning differential-drive non-holonomic kinematics and 3D rigid-body inertia tensors, dual-source continuous-discrete Extended Kalman Filter (EKF) sensor fusion, Adaptive Monte Carlo Localization (AMCL) state sampling, Ceres-based pose-graph SLAM optimization, potential field costmap inflation, Euclidean Distance Transform (EDT) topological roadmap extraction, Model Predictive Path Integral (MPPI) stochastic control, C^1-continuous cubic Bézier reactive evasion trajectories, wavefront frontier exploration, pinhole camera projection geometry, and affine Canvas web dashboard coordinate transformations.

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1 Robot Kinematics, Dynamics & Inertial Physics

The AMR is modeled as a two-wheel differential drive ground vehicle stabilized by front and rear passive low-friction spherical casters, operating in the planar special Euclidean group $SE(2)$.

1.1 Differential Drive Forward Kinematics

Let $(x, y, \theta)^T \in SE(2)$ denote the robot pose in the global inertial reference frame. The physical platform parameters derived from the unified robot description format (`warehouse_agv.urdf`) are:

- Wheel radius: $r = 0.025$ m (nominal wheel diameter 50 mm)
- Wheel track baseline: $L = 0.26$ m (wheel centers positioned at $y = \pm 0.13$ m)
- Chassis footprint radius: $R_{\text{base}} = 0.11$ m (chassis cylinder diameter 220 mm)
- Total mobile base mass: $M = 10.0$ kg

Given individual left and right wheel angular rotational velocities $\omega_L, \omega_R \in \mathbb{R}$, the forward translational velocity v and yaw rate ω in the robot body frame are defined by:

$$v = \frac{r}{2}(\omega_R + \omega_L) \tag{1}$$

$$\omega = \frac{r}{L}(\omega_R - \omega_L) \tag{2}$$

In matrix mapping form:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} \frac{r}{2} & \frac{r}{2} \\ \frac{r}{L} & -\frac{r}{L} \end{bmatrix} \begin{bmatrix} \omega_R \\ \omega_L \end{bmatrix} \tag{3}$$

The non-holonomic continuous-time state-space kinematic model in the global inertial frame is:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos\theta(t) & 0 \\ \sin\theta(t) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v(t) \\ \omega(t) \end{bmatrix} \tag{4}$$

Subject to the non-holonomic zero lateral slip velocity constraint:

$$-\dot{x}(t)\sin\theta(t) + \dot{y}(t)\cos\theta(t) = 0 \tag{5}$$

1.2 Inverse Kinematics

Given an arbitrary commanded velocity twist $(v, \omega)^T$ issued by the trajectory controller, the required individual wheel angular velocities are calculated by inverting the kinematic matrix:

$$\omega_R = \frac{v + \frac{L}{2}\omega}{r} = \frac{2v + L\omega}{2r} \tag{6}$$

$$\omega_L = \frac{v - \frac{L}{2}\omega}{r} = \frac{2v - L\omega}{2r} \tag{7}$$

1.3 Exact Circular Arc Odometry Dead-Reckoning

Over discrete sampling period Δt, the left and right wheel displacement increments are $\Delta s_L = r\Delta\theta_L$ and $\Delta s_R = r\Delta\theta_R$. The net chord displacement Δs and rotational change $\Delta\theta$ are:

$$\Delta s = \frac{\Delta s_R + \Delta s_L}{2}, \quad \Delta\theta = \frac{\Delta s_R - \Delta s_L}{L} \tag{8}$$

When $\Delta\theta \neq 0$, the robot moves along an instantaneous circular arc of radius $R_{\text{arc}} = \frac{\Delta s}{\Delta\theta}$. The exact closed-form integral over $[t_k, t_{k+1}]$ is:

$$x_{k+1} = x_k + \Delta s \cdot \frac{\sin(\Delta\theta)}{\Delta\theta} \cos\left(\theta_k + \frac{\Delta\theta}{2}\right) \tag{9}$$

$$y_{k+1} = y_k + \Delta s \cdot \frac{\sin(\Delta\theta)}{\Delta\theta} \sin\left(\theta_k + \frac{\Delta\theta}{2}\right) \tag{10}$$

$$\theta_{k+1} = \theta_k + \Delta\theta \tag{11}$$

For quasi-straight motion where $|\Delta\theta| < 10^{-6}$, the third-order Taylor expansion replaces the division by zero:

$$\frac{\sin(\Delta\theta)}{\Delta\theta} \approx 1 - \frac{(\Delta\theta)^2}{6} + \mathcal{O}((\Delta\theta)^4) \tag{12}$$

$$x_{k+1} = x_k + \Delta s \cos\left(\theta_k + \frac{\Delta\theta}{2}\right), \quad y_{k+1} = y_k + \Delta s \sin\left(\theta_k + \frac{\Delta\theta}{2}\right) \tag{13}$$

1.4 Rigid-Body Mass & Moment of Inertia Tensors

The AMR's dynamic response and Gazebo physics simulation rely on 3D spatial inertia tensors:

1. **Cylindrical Chassis Body:** Radius $R = 0.11$ m, height $H = 0.08$ m, total mass $m = 10.0$ kg:

$$I_{xx} = I_{yy} = \frac{1}{12}m(3R^2 + H^2) = \frac{10}{12}(3(0.11)^2 + (0.08)^2) = 0.03558 \text{ kg} \cdot \text{m}^2 \tag{14}$$

$$I_{zz} = \frac{1}{2}mR^2 = \frac{1}{2}(10)(0.11)^2 = 0.06050 \text{ kg} \cdot \text{m}^2 \tag{15}$$

2. **Solid Drive Wheels:** Cylinders with mass $m_w = 0.5$ kg, radius $r = 0.025$ m, width $w = 0.02$ m:

$$I_{zz} = \frac{1}{2}m_w r^2 = \frac{1}{2}(0.5)(0.025)^2 = 1.5625 \times 10^{-4} \text{ kg} \cdot \text{m}^2 \tag{16}$$

$$I_{xx} = I_{yy} = \frac{1}{12}m_w(3r^2 + w^2) = \frac{0.5}{12}(3(0.025)^2 + (0.02)^2) = 9.479 \times 10^{-5} \text{ kg} \cdot \text{m}^2 \tag{17}$$

2 State Estimation & Sensor Fusion

2.1 Extended Kalman Filter (EKF Fusion)

The state estimation pipeline combines high-frequency wheel encoder dead-reckoning (50 Hz) and IMU angular rate / linear acceleration (50 Hz) using a 15-state non-linear continuous-discrete Extended Kalman Filter (`robot_localization`).

The full continuous state vector is:

$$\mathbf{x} = [x, y, z, \phi, \theta, \psi, \dot{x}, \dot{y}, \dot{z}, \dot{\phi}, \dot{\theta}, \dot{\psi}, \ddot{x}, \ddot{y}, \ddot{z}]^T \in \mathbb{R}^{15} \tag{18}$$

For planar ground navigation (`two_d_mode: true`), the operational state vector contracts to:

$$\mathbf{x}_{2D} = [x, y, \psi, \dot{x}, \dot{y}, \dot{\psi}, \ddot{x}, \ddot{y}]^T \in \mathbb{R}^8 \tag{19}$$

2.1.1 Continuous-Discrete Time State Prediction

The continuous-time process evolution model with zero-mean Gaussian white process noise $\mathbf{w}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_c)$ is:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)) + \mathbf{w}(t) \tag{20}$$

For a sample period Δt, the discrete-time forward state propagation is:

$$\hat{\mathbf{x}}_{k|k-1} = \hat{\mathbf{x}}_{k-1|k-1} + \int_{t_{k-1}}^{t_k} \mathbf{f}(\hat{\mathbf{x}}(\tau))d\tau \tag{21}$$

The state transition Jacobian $\mathbf{F}_k = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\big|_{\hat{\mathbf{x}}_{k-1|k-1}}$ yields the discrete covariance prediction:

$$\mathbf{P}_{k|k-1} = \mathbf{\Phi}_k \mathbf{P}_{k-1|k-1} \mathbf{\Phi}_k^T + \mathbf{Q}_k, \quad \mathbf{\Phi}_k \approx \mathbf{I} + \mathbf{F}_k \Delta t \tag{22}$$

2.1.2 Discrete Measurement Updates

When an observation $\mathbf{z}_k \in \mathbb{R}^m$ arrives from wheel odometry or IMU with noise covariance $\mathbf{R}_k$:

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}) \quad \text{(Innovation Residual)} \tag{23}$$

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k \quad \text{(Innovation Covariance)} \tag{24}$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T \mathbf{S}_k^{-1} \quad \text{(Optimal Kalman Gain)} \tag{25}$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \mathbf{y}_k \quad \text{(State Update)} \tag{26}$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)^T + \mathbf{K}_k \mathbf{R}_k \mathbf{K}_k^T \quad \text{(Joseph Form Covariance)} \tag{27}$$

2.2 Adaptive Monte Carlo Localization (AMCL)

AMCL estimates the robot global pose $S_t = (x, y, \theta)^T$ using a particle filter $\{S_t^{[m]}, w_t^{[m]}\}_{m=1}^M$ conditioned on laser range scans z_t and map m.

2.2.1 Sampling Odometry Motion Model

Given dead-reckoning odometry displacement $\bar{x}_{t-1} = (\bar{x}, \bar{y}, \bar{\theta})^T$ and $\bar{x}_t = (\bar{x}', \bar{y}', \bar{\theta}')^T$, the relative motion parameters are:

$$\delta_{\text{rot1}} = \text{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta} \tag{28}$$

$$\delta_{\text{trans}} = \sqrt{(\bar{x}' - \bar{x})^2 + (\bar{y}' - \bar{y})^2} \tag{29}$$

$$\delta_{\text{rot2}} = \bar{\theta}' - \bar{\theta} - \delta_{\text{rot1}} \tag{30}$$

Noisy perturbations are sampled using error parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4$:

$$\hat{\delta}_{\text{rot1}} = \delta_{\text{rot1}} - \text{sample}(\alpha_1 \delta_{\text{rot1}}^2 + \alpha_2 \delta_{\text{trans}}^2) \tag{31}$$

$$\hat{\delta}_{\text{trans}} = \delta_{\text{trans}} - \text{sample}(\alpha_3 \delta_{\text{trans}}^2 + \alpha_4 (\delta_{\text{rot1}}^2 + \delta_{\text{rot2}}^2)) \tag{32}$$

$$\hat{\delta}_{\text{rot2}} = \delta_{\text{rot2}} - \text{sample}(\alpha_1 \delta_{\text{rot2}}^2 + \alpha_2 \delta_{\text{trans}}^2) \tag{33}$$

2.2.2 Likelihood Field Sensor Model

Each LiDAR beam endpoint in global coordinates:

$$x_{z_t^k} = x + x_{\text{laser}} \cos\theta - y_{\text{laser}} \sin\theta + z_t^k \cos(\theta + \theta_{\text{laser}} + \phi_k) \tag{34}$$

$$y_{z_t^k} = y + x_{\text{laser}} \sin\theta + y_{\text{laser}} \cos\theta + z_t^k \sin(\theta + \theta_{\text{laser}} + \phi_k) \tag{35}$$

The likelihood probability mixes zero-delay Gaussian hit probability and random noise:

$$p(z_t^k | S_t, m) = z_{\text{hit}} \cdot \frac{1}{\sqrt{2\pi\sigma_{\text{hit}}^2}} \exp\left(-\frac{\text{dist}(x_{z_t^k}, y_{z_t^k}, m)^2}{2\sigma_{\text{hit}}^2}\right) + \frac{z_{\text{rand}}}{z_{\max}} \tag{36}$$

2.2.3 Kullback-Leibler Divergence (KLD) Particle Bounds

The dynamic particle count M required to keep the approximation error bounded by $\epsilon = 0.05$ with probability $1 - \delta = 0.99$ across k occupied bins is:

$$M \approx \frac{k-1}{2\epsilon} \left(1 - \frac{2}{9(k-1)} + \sqrt{\frac{2}{9(k-1)}} z_{1-\delta} \right)^3 \tag{37}$$

Clamped to the configured operational range: $M_{\min} = 500 \leq M \leq M_{\max} = 2000$.

3 SLAM & Pose-Graph Optimization

`slam_toolbox` optimizes robot trajectory poses $\mathbf{x} = \{x_1, \ldots, x_N\}$ and laser scan matches using non-linear least squares.

3.1 Correlative Scan Matching

Given current scan points $P = \{p_i\}$ and previous local map $\mathcal{M}$, the optimal rigid transform $\hat{\xi} = (\hat{t}_x, \hat{t}_y, \hat{\theta})^T \in SE(2)$ maximizes the correlation score:

$$\hat{\xi} = \arg\max_{\xi} \sum_{i=1}^{|P|} \mathcal{M}(T(\xi) \cdot p_i) \tag{38}$$

3.2 Ceres Pose-Graph Non-Linear Least Squares

The global trajectory cost functional over odometry edges $\mathcal{E}_{\text{odom}}$ and loop-closure constraints $\mathcal{E}_{\text{loop}}$ is:

$$F(\mathbf{x}) = \sum_{(i,j) \in \mathcal{E}} \mathbf{e}_{ij}(\mathbf{x}_i, \mathbf{x}_j)^T \mathbf{\Omega}_{ij} \mathbf{e}_{ij}(\mathbf{x}_i, \mathbf{x}_j) \tag{39}$$

Where the $SE(2)$ relative transformation error is:

$$\mathbf{e}_{ij}(\mathbf{x}_i, \mathbf{x}_j) = \begin{bmatrix} \mathbf{R}_i^T(\mathbf{t}_j - \mathbf{t}_i) - \mathbf{t}_{ij}^{\text{meas}} \\ \text{wrap2pi}(\theta_j - \theta_i - \Delta\theta_{ij}^{\text{meas}}) \end{bmatrix} \tag{40}$$

The non-linear system is iteratively solved using the Levenberg-Marquardt trust-region algorithm:

$$\left(\mathbf{J}^T \mathbf{\Omega} \mathbf{J} + \lambda \, \text{diag}(\mathbf{J}^T \mathbf{\Omega} \mathbf{J}) \right) \Delta \mathbf{x} = -\mathbf{J}^T \mathbf{\Omega} \mathbf{e} \tag{41}$$

4 Costmap Inflation & Potential Fields

The 2D Nav2 Costmap transforms binary obstacle geometry into continuous navigation risks.

$$C(d) = \begin{cases} 254 & \text{if } d = 0 \text{ (Lethal Obstacle)} \\ 253 & \text{if } 0 < d \leq R_{\text{inscribed}} \text{ (Inscribed Footprint Danger)} \\ \lfloor 252 \cdot \exp(-k \cdot (d - R_{\text{inscribed}})) \rfloor & \text{if } R_{\text{inscribed}} < d \leq R_{\text{inflation}} \\ 0 & \text{if } d > R_{\text{inflation}} \text{ (Free Navigable Space)} \end{cases} \tag{42}$$

Platform parameters:

- $R_{\text{inscribed}} = 0.13$ m (enclosing footprint radius $R_{\text{base}} = 0.11$ m plus 20 mm margin)
- $R_{\text{inflation}} = 0.25$ m (smooth repulsion buffer)
- Inflation decay scale: $k = 3.0$ m^{-1}

5 Topological Roadmap & Graph Theory

5.1 Euclidean Distance Transform (EDT)

Given binary grid occupancy $M(u, v) \in \{0, 1\}$, the distance to the nearest obstacle boundary is computed via the continuous transform:

$$D(u, v) = \min_{(u', v') : M(u', v') = 1} \sqrt{(u - u')^2 + (v - v')^2} \cdot \text{resolution} \tag{43}$$

5.2 Swept-Circle Clearance Verification

For an edge connecting topological nodes $\mathbf{p}_A$ and $\mathbf{p}_B$, collision-free safety is guaranteed by sampling along the parameterized chord:

$$\mathbf{p}(s) = (1-s)\mathbf{p}_A + s\mathbf{p}_B, \quad s \in [0,1] \tag{44}$$

$$\min_{s\in[0,1]} D(\mathbf{p}(s)) \geq R_{\text{robot_margin}} = 0.18 \text{ m} \tag{45}$$

5.3 Dijkstra Shortest Path Routing

The global planner extracts paths over graph $G = (V, E)$ by minimizing edge traversal distances:

$$w(u,v) = \|\mathbf{p}_u - \mathbf{p}_v\|_2 = \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2} \tag{46}$$

$$\text{dist}[v] = \min_{(u,v)\in E} (\text{dist}[u] + w(u,v)) \tag{47}$$

6 Local Motion Planning: Model Predictive Path Integral (MPPI)

The AMR uses an Model Predictive Path Integral (MPPI) controller for real-time trajectory optimization under dynamic obstacles.

6.1 Stochastic Trajectory Rollouts

At control cycle t_k, $K = 80$ stochastic candidate trajectories are sampled over a prediction horizon $T = 15$ steps ($\Delta t = 0.10$ s, total lookahead horizon 1.5 s):

$$v_k^{(i)} = \bar{v}_k + \epsilon_{v,k}^{(i)}, \quad \epsilon_v \sim \mathcal{N}(0, \sigma_v^2), \quad \sigma_v = 0.08 \text{ m/s} \tag{48}$$

$$\omega_k^{(i)} = \bar{\omega}_k + \epsilon_{\omega,k}^{(i)}, \quad \epsilon_\omega \sim \mathcal{N}(0, \sigma_\omega^2), \quad \sigma_\omega = 0.35 \text{ rad/s} \tag{49}$$

Each trajectory is propagated forward using non-holonomic unicycle integration:

$$x_{t+1}^{(i)} = x_t^{(i)} + v_t^{(i)} \cos(\theta_t^{(i)}) \Delta t \tag{50}$$

$$y_{t+1}^{(i)} = y_t^{(i)} + v_t^{(i)} \sin(\theta_t^{(i)}) \Delta t \tag{51}$$

$$\theta_{t+1}^{(i)} = \theta_t^{(i)} + \omega_t^{(i)} \Delta t \tag{52}$$

6.2 Composite Cost Functional

The total cost $S(\tau^{(i)})$ of rollout $\tau^{(i)}$ balances six objective criteria:

$$S(\tau^{(i)}) = J_{\text{term}} + J_{\text{heading}} + J_{\text{cte}} + J_{\text{repulsion}} + J_{\text{dyn_coll}} + J_{\text{effort}} \tag{53}$$

1. **Terminal Waypoint Distance Cost:**

$$J_{\text{term}} = w_{\text{dist}} \cdot \left\| \mathbf{p}_T^{(i)} - \mathbf{p}_{\text{target}} \right\|_2, \quad w_{\text{dist}} = 4.0 \tag{54}$$

2. **Horizon Heading Alignment Cost:**

$$J_{\text{heading}} = w_{\text{heading}} \cdot \left| \text{wrap2pi}\left(\theta_T^{(i)} - \text{atan2}(y_{\text{tgt}} - y_T^{(i)}, x_{\text{tgt}} - x_T^{(i)}) \right) \right|, \quad w_{\text{heading}} = 3.0 \tag{55}$$

3. **Cross-Track Error (CTE) Cost:**

$$J_{\text{cte}} = w_{\text{cte}} \sum_{t=1}^{T} d_\perp(\mathbf{p}_t^{(i)}, \text{Path Segment}), \quad w_{\text{cte}} = 6.0 \text{ (nominal)}, \quad 2.0 \text{ (evasion)} \tag{56}$$

4. **Static Obstacle Distance Repulsion:**

$$J_{\text{repulsion}} = w_{\text{rep}} \sum_{t=1}^{T} \max\left(0, d_{\text{safe}} - \text{dist}(\mathbf{p}_t^{(i)}, \text{Obstacles})\right)^2 \tag{57}$$

5. **Predictive Dynamic Obstacle Collision Penalty:**

$$J_{\text{dyn_coll}} = \sum_{t=1}^{T} \sum_{m=1}^{M_{\text{obs}}} \begin{cases} w_{\text{coll}} & \text{if } \|\mathbf{p}_t^{(i)} - \hat{\mathbf{p}}_{m,t}\|_2 < R_{\text{coll}} = 0.28 \text{ m} \\ 0 & \text{otherwise} \end{cases} \tag{58}$$

where $w_{\text{coll}} = 5000.0$.

6.3 Softmax Optimal Action Selection

The optimal command vector $\mathbf{u}_t^* = (v_t^*, \omega_t^*)^T$ is evaluated as the importance-weighted expectation:

$$\beta = \min_{i=1,\dots,K} S(\tau^{(i)}) \tag{59}$$

$$w^{(i)} = \frac{\exp\left(-\frac{1}{\lambda}(S(\tau^{(i)}) - \beta)\right)}{\sum_{j=1}^{K} \exp\left(-\frac{1}{\lambda}(S(\tau^{(j)}) - \beta)\right)}, \quad \lambda = 0.50 \tag{60}$$

$$\mathbf{u}_t^* = \sum_{i=1}^{K} w^{(i)} \mathbf{u}_t^{(i)} \tag{61}$$

6.4 Slew-Rate Limiting & EMA Velocity Smoothing

To ensure actuator continuity and eliminate jerking motion:

$$v_{\text{slew}} = \begin{cases} \min(v_t^*, v_{t-1} + a_{\text{accel}}\Delta t) & \text{if } v_t^* \geq v_{t-1} \\ \max(v_t^*, v_{t-1} - a_{\text{decel}}\Delta t) & \text{if } v_t^* < v_{t-1} \end{cases} \tag{62}$$

$$\omega_{\text{slew}} = \text{clamp}\left(\omega_t^*, \omega_{t-1} - \alpha_{\text{accel}}\Delta t, \omega_{t-1} + \alpha_{\text{accel}}\Delta t\right) \tag{63}$$

Where $a_{\text{accel}} = 0.60$ m/s^2, $a_{\text{decel}} = 0.90$ m/s^2, and $\alpha_{\text{accel}} = 1.60$ rad/s^2.

Finally, an Exponential Moving Average (EMA) filter is applied:

$$v_{\text{final}} = \alpha_v \cdot v_{\text{slew}} + (1 - \alpha_v) \cdot v_{\text{prev}}, \quad \alpha_v = 0.70 \tag{64}$$

$$\omega_{\text{final}} = \alpha_\omega \cdot \omega_{\text{slew}} + (1 - \alpha_\omega) \cdot \omega_{\text{prev}}, \quad \alpha_\omega = 0.70 \tag{65}$$

7 Dynamic Obstacle Tracking & Evasion Geometry

7.1 Clustering and Velocity Estimation

LiDAR scan endpoints are clustered using Euclidean radius filtering ($d_{\text{cluster}} = 0.35$ m). The cluster centroid is:

$$\bar{\mathbf{p}}_k = \frac{1}{|C|} \sum_{p_i \in C} p_i \tag{66}$$

Obstacle velocity is estimated by finite differences over timestamp separation Δt:

$$\mathbf{v}_{\text{obs}} = \frac{\bar{\mathbf{p}}_k - \bar{\mathbf{p}}_{k-1}}{\Delta t} \tag{67}$$

7.2 Vector Cross-Product Swerve Decision

To commit to a consistent evasion corridor (left versus right avoidance):

$$\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{robot}} - \mathbf{v}_{\text{obs}} \tag{68}$$

$$\mathbf{r}_{\text{obs}} = \mathbf{p}_{\text{obs}} - \mathbf{p}_{\text{robot}} \tag{69}$$

The 2D scalar cross-product defines the evasion commitment sign:

$$\zeta = \mathbf{v}_{\text{rel}} \times \mathbf{r}_{\text{obs}} = v_{\text{rel},x} r_{\text{obs},y} - v_{\text{rel},y} r_{\text{obs},x} \tag{70}$$

$$\text{Side} = \begin{cases} +1 \text{ (Left Swerve)} & \text{if } \zeta > 0 \\ -1 \text{ (Right Swerve)} & \text{if } \zeta \leq 0 \end{cases} \tag{71}$$

7.3 C^1-Continuous Cubic Bézier Detour Trajectory

The evasion path is synthesized as a parametric cubic polynomial curve:

$$\mathbf{B}(u) = (1-u)^3\mathbf{P}_0 + 3(1-u)^2u\mathbf{P}_1 + 3(1-u)u^2\mathbf{P}_2 + u^3\mathbf{P}_3, \quad u \in [0,1] \tag{72}$$

With control point anchors:

- $\mathbf{P}_0 = \mathbf{p}_{\text{start}}$
- $\mathbf{P}_1 = \mathbf{P}_0 + \kappa \cdot \mathbf{t}_{\text{start}}$ (satisfying velocity alignment $\mathbf{B}'(0) = 3(\mathbf{P}_1 - \mathbf{P}_0)$)
- $\mathbf{P}_2 = \mathbf{P}_3 - \kappa \cdot \mathbf{t}_{\text{end}}$
- $\mathbf{P}_3 = \mathbf{p}_{\text{rejoin}}$

8 Autonomous Exploration & Multi-Robot Map Merging

8.1 Wavefront Frontier Detection (WFD)

A frontier point $\mathbf{x}_f$ satisfies:

$$M(\mathbf{x}_f) = \text{Free} \quad \wedge \quad \exists \mathbf{x}_{\text{nbr}} \in \mathcal{N}_8(\mathbf{x}_f) \text{ s.t. } M(\mathbf{x}_{\text{nbr}}) = \text{Unknown} \tag{73}$$

8.2 Exploration Utility Function

Candidate frontier clusters are scored by trading off expected information gain against navigation cost:

$$U(f) = w_{\text{gain}} \cdot \text{Area}(f) - w_{\text{cost}} \cdot \text{Cost}_{\text{Dijkstra}}(\mathbf{p}_{\text{robot}}, \text{centroid}(f)) \tag{74}$$

8.3 Multi-Robot $SE(2)$ Map Merging

Given maps $\mathcal{M}_A$ and $\mathcal{M}_B$, the rigid transform $(\mathbf{R}, \mathbf{t})$ aligns features via RANSAC:

$$\mathbf{p}_i^A \approx \mathbf{R}\mathbf{p}_i^B + \mathbf{t} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_i^B \\ y_i^B \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \tag{75}$$

Merged cell values are fused in log-odds space:

$$L(m_{x,y}) = \ln\left(\frac{P(m_{x,y})}{1 - P(m_{x,y})}\right) \tag{76}$$

$$L_{\text{merged}} = L_A(x,y) + L_B(T(x,y)) - L_0 \tag{77}$$

9 Computer Vision Geometry

9.1 Pinhole Camera Intrinsic Projection Matrix

A 3D spatial landmark point $\mathbf{P}_c = (X_c, Y_c, Z_c)^T$ in camera coordinates projects to image plane pixel coordinates $(u, v)^T$:

$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}\mathbf{P}_c = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} \tag{78}$$

$$u = f_x\frac{X_c}{Z_c} + c_x, \quad v = f_y\frac{Y_c}{Z_c} + c_y \tag{79}$$

9.2 Normalized Pixel Offset & Depth Proxy

The horizontal offset angle ψ for detected objects is:

$$\Delta u = \frac{u_{\text{bbox_center}} - c_x}{W/2} \in [-1, 1] \tag{80}$$

$$\psi = \arctan\left(\frac{\Delta u \cdot (W/2)}{f_x}\right) \tag{81}$$

Distance is approximated using known physical height H_{obj}:

$$Z_{\text{est}} = \frac{f_y \cdot H_{\text{obj}}}{h_{\text{bbox}}} \tag{82}$$

10 Web Dashboard Coordinate Transformations

The HTML5 Canvas dashboard maps continuous ROS world coordinates (x_w, y_w) to display pixels (p_x, p_y):

$$p_x = \frac{x_w - x_0}{\text{resolution}} \tag{83}$$

$$p_y = H_{\text{map}} - \frac{y_w - y_0}{\text{resolution}} \tag{84}$$

With user zoom scale s and pan translation (T_x, T_y), screen coordinates (S_x, S_y) are:

$$\begin{bmatrix} S_x \\ S_y \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & T_x \\ 0 & s & T_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ 1 \end{bmatrix} \tag{85}$$

Smoothed robot orientation and position render via 60 FPS Linear Interpolation (LERP):

$$x(t) = (1-\gamma)x(t-1) + \gamma x_{\text{new}}, \quad \gamma = 0.25 \tag{86}$$

$$\theta(t) = \theta(t-1) + \gamma \cdot \operatorname{atan2}\left(\sin(\theta_{\text{new}} - \theta(t-1)), \cos(\theta_{\text{new}} - \theta(t-1))\right) \tag{87}$$

11 System Constants Reference Table

Parameter Name	**Symbol**	**Value & Units**	**Subsystem Reference**
Wheel Radius	r	0.025 m	URDF Kinematics
Wheel Baseline Track	L	0.26 m	URDF Kinematics
Chassis Mass	M	10.0 kg	Rigid Body Dynamics
Chassis Radius	R_{base}	0.11 m	Footprint Clearance
Inscribed Radius	$R_{\text{inscribed}}$	0.13 m	Nav2 Costmap
Inflation Radius	$R_{\text{inflation}}$	0.25 m	Nav2 Potential Field
Costmap Decay Rate	k	3.0 m^{-1}	Exponential Costmap
Rollout Trajectories	K	80	MPPI Controller
Prediction Steps	T	15	MPPI Controller
Time Step Horizon	Δt	0.10 s	MPPI Controller
Temperature Parameter	λ	0.50	MPPI Softmax
Linear Acceleration Limit	a_{accel}	0.60 m/s^2	Slew Rate Limiter
Linear Deceleration Limit	a_{decel}	0.90 m/s^2	Slew Rate Limiter
Angular Acceleration Limit	α_{accel}	1.60 rad/s^2	Slew Rate Limiter
Velocity EMA Filter Alpha	α_v, α_ω	0.70	Low-Pass Filter
Collision Risk Radius	R_{coll}	0.28 m	Dynamic Collision Avoidance
Min Particles	M_{min}	500	AMCL Particle Filter
Max Particles	M_{max}	2000	AMCL Particle Filter
Map Resolution	res	0.05 m/pixel	SLAM Occupancy Grid

End of Autonomous Mobile Robot (AMR) Mathematical Formulations Report.